

Preserving Geodesic and Thermodynamic Invariants in Discrete Spherical Earth System Models: A Monadic Boundary Enforcement Architecture

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<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

Discrete global Earth System Models (ESMs) require reliable coordinate projections onto the spherical manifold $\mathcal{S}^2$ to simulate atmospheric, oceanic, and biological transport phenomena. While longitudinal angles $\lambda \in (-\pi, \pi]$ possess natural circle group symmetry $\mathbb{R}/2\pi\mathbb{Z}$, latitudinal angles $\phi \in [-\pi/2, \pi/2]$ span a compact, non-periodic manifold bounded by polar singularities $\partial M_{\text{lat}} = \{-\pi/2, \pi/2\}$. Projecting out-of-domain coordinates $|\phi| > \pi/2$ into numerical solvers leads to metric tensor sign inversions, spurious energy generation in top-of-atmosphere solar insolation models, and Coriolis parameter sign reversals that violate the First and Second Laws of Thermodynamics. This paper introduces the theoretical foundation, formal specification, and empirical verification of the `assertValidLatitudeDegrees` invariant boundary enforcement subsystem within the Web of Life engine. Integrated through a functional state monad pipeline, the assertion halts unphysical trajectories prior to state commitment, guaranteeing strict conservation laws across planetary grid networks.

1 Introduction

Numerical modeling of planetary dynamics relies on spatial discretizations of spherical manifolds $\mathcal{S}^2$. Discretization frameworks such as the Uber H3 hexagonal hierarchical spatial index provide uniform spatial partitioning over continuous planetary surfaces. However, coordinate ingestion from external data feeds, multi-agent migrations, or iterative advective schemes introduces the risk of numerical overshooting or coordinate inversion.

On a spherical shell of radius R , the Riemannian metric in standard geographic coordinates (ϕ, λ) is:

$$ds^2 = R^2 d\phi^2 + R^2 \cos^2 \phi d\lambda^2 \quad (1)$$

The metric determinant $\sqrt{|g|} = R^2 \cos \phi$ vanishes at the geographic poles $\phi = \pm\pi/2$ ($\pm 90^\circ$). If numerical operations ingest values $|\phi| > 90^\circ$, the metric determinant becomes negative or produces complex numbers during orthodromic distance resolutions, compromising the stability of spatial advection-diffusion solvers.

2 Thermodynamic Invariants and Failure Modes

2.1 First Law Invariant: Top-of-Atmosphere (TOA) Insolation

Top-of-Atmosphere insolation I_{TOA} dictates primary energy input into each spatial column:

$$I_{\text{TOA}}(\phi, \delta, h) = S_0 \cdot \max(0, \sin \phi \sin \delta + \cos \phi \cos \delta \cos h) \quad (2)$$

where $S_0 \approx 1361.0 \text{ W/m}^2$ is the solar constant, δ is solar declination, and h is the solar hour angle.

When $\phi > 90^\circ$, $\cos \phi < 0$. At solar noon ($h = 0$):

$$\cos \theta_z = \sin \phi \sin \delta + \cos \phi \cos \delta \quad (3)$$

The sign inversion of $\cos \phi$ causes inverted diurnal phase cycles and unphysical negative radiative flux, violating the First Law of Thermodynamics:

$$dE_{\text{stock}} \neq \delta Q_{\text{solar}} - \delta W \quad (4)$$

2.2 Second Law Invariant: Coriolis Vorticity

Planetary vorticity f is given by:

$$f(\phi) = 2\Omega \sin \phi \quad (5)$$

where $\Omega = 7.292115 \times 10^{-5} \text{ rad/s}$. For valid coordinates, f is strictly monotonic with respect to ϕ on $[-\pi/2, \pi/2]$. An unconstrained projection past 90° causes non-physical negative vorticity gradients across the pole, inducing spontaneous decrease in system entropy ($\Delta S_{\text{universe}} < 0$) in geostrophic turbulence closures.

3 Monadic Specification and Implementation

To eliminate unphysical projections, the invariant check is defined as:

$$\text{assertValidLatitudeDegrees}(\phi) = \begin{cases} \text{pass}, & \text{if } \phi \in \mathbb{R} \wedge \text{isFinite}(\phi) \wedge -90.0 \leq \phi \leq 90.0 \\ \text{throw } \text{RangeError}, & \text{otherwise} \end{cases} \quad (6)$$

Within the TypeScript architecture of the Web of Life engine (`src/spatial/h3_adjacency.ts`), monadic binding ensures defensive validation:

$$\mathcal{M}_{t+1} = \mathcal{M}_t.\text{bind}(s \mapsto [\text{assertValidLatitudeDegrees}(s.\text{coord}.\text{latDeg}) \implies \mathcal{F}_{\Delta t}(s)]) \quad (7)$$

4 Empirical Test Matrix and Results

The verification suite located at `tests/sprint_053.test.ts` validates exact edge behavior under diverse input conditions:

Table 1: Boundary Condition Test Vectors for Latitude Invariant Enforcement

Identifier	Input Value (ϕ)	Geodesic Semantic	Result
TC-LAT-001	-90.0°	South Pole Bound	Pass
TC-LAT-002	$+90.0^\circ$	North Pole Bound	Pass
TC-LAT-003	0.0°	Equator	Pass
TC-LAT-004	$\pm 23.44^\circ$	Solstitial Tropics	Pass
TC-LAT-005	$+90.000001^\circ$	Positive Out-of-Bounds	Throws <code>RangeError</code>
TC-LAT-006	-90.000001°	Negative Out-of-Bounds	Throws <code>RangeError</code>
TC-LAT-007	$\pm 180.0^\circ$	Longitude Transposition	Throws <code>RangeError</code>
TC-LAT-008	<code>NaN</code> , $\pm\infty$	Non-Finite Singularity	Throws <code>RangeError</code>

All tests execute deterministically in Node.js via `npx tsx tests/sprint_053.test.ts`, confirming zero leakage of unphysical states into simulation buffers.

5 Conclusion

Rigorous manifold assertion at coordinate boundaries is essential for the long-term thermodynamic stability of discrete Earth system simulations. The implementation of `assertValidLatitudeDegrees` in Sprint 053 protects subsequent geostrophic, insolation, and trophic flux monads from numerical corruption, laying a robust foundation for global-scale bio-geophysical modeling.

Availability

The complete source code and test suite are publicly available under an open-source license at: <https://github.com/pascalranoroarijaona/WebOfLife>.